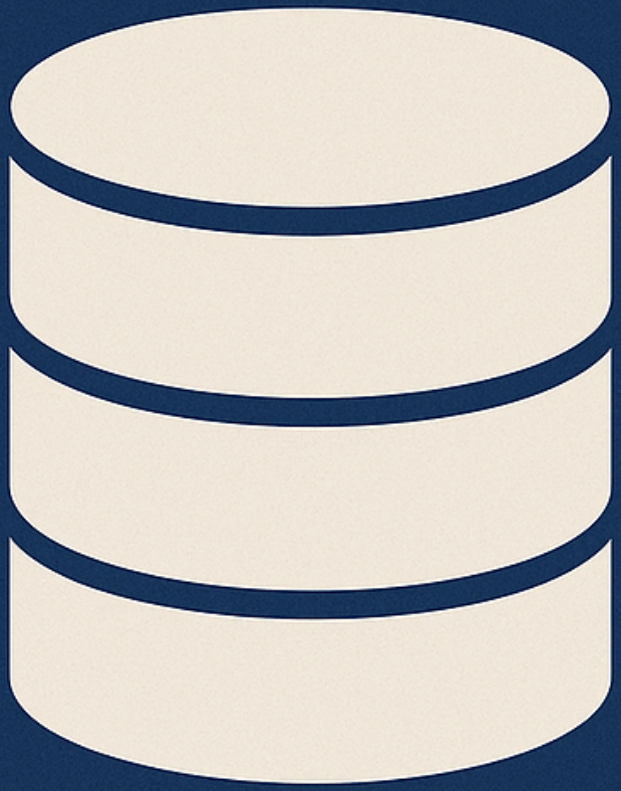
SQL

Class starts
@ 9:10pm





Lecture 11: SQL Joins Continued

#Agenda

1) Recap + Questions
2) Full Outer Join
3) Joining multiple tables
4) Self Join
5) Cross Join

Dataset: Global Tech Solution Dataset

Types

1) Inner Join
2) Left Join
3) Right Join

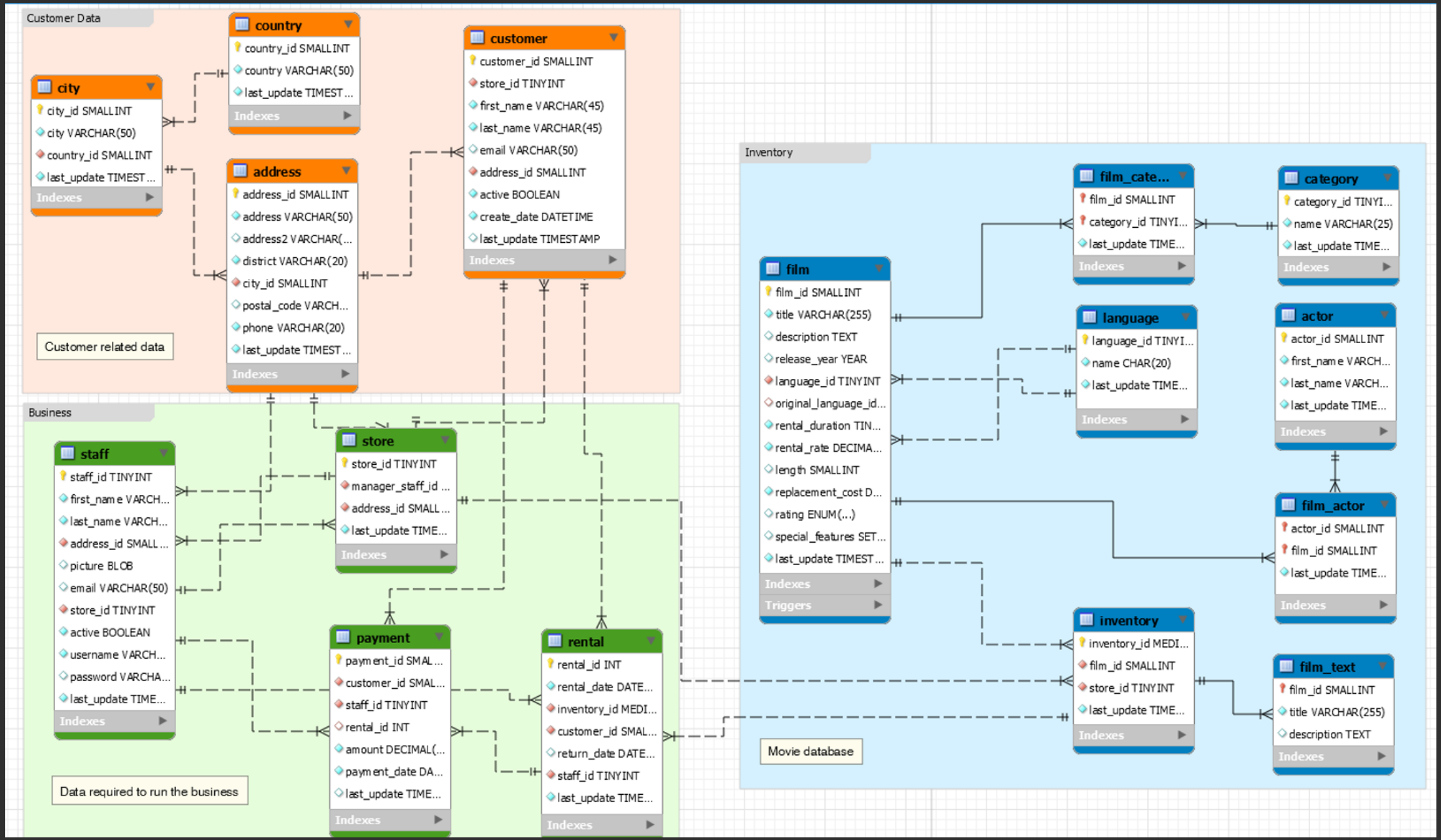

Customer Data
country
country_id SMALLINT
country VARCHAR(50)
last_update TIMEST...
Indexes
customer
customer_id SMALLINT
store_id TINYINT
first_name VARCHAR(45)
last_name VARCHAR(45)
email VARCHAR(50)
address_id SMALLINT
active BOOLEAN
create_date DATETIME
last_update TIMESTAMP
Indexes
city
city_id SMALLINT
city VARCHAR(50)
country_id SMALLINT
last_update TIMEST...
Indexes
address
address_id SMALLINT
address VARCHAR(50)
address2 VARCHAR(...
district VARCHAR(20)
city_id SMALLINT
postal_code VARCH...
phone VARCHAR(20)
last_update TIMEST...
Indexes
Customer related data
Business
staff
staff_id TINYINT
first_name VARCH...
last_name VARCH...
address_id SMALL...
picture BLOB
email VARCHAR(50)
store_id TINYINT
active BOOLEAN
username VARCH...
password VARCHA...
last_update TIME...
Indexes
store
store_id TINYINT
manager_staff_id ...
address_id SMALL...
last_update TIME...
Indexes
payment
payment_id SMAL...
customer_id SMAL...
staff_id TINYINT
rental_id INT
amount DECIMAL(...
payment_date DA...
last_update TIME...
Indexes
rental
rental_id INT
rental_date DATE...
inventory_id MEDI...
customer_id SMAL...
return_date DATE...
staff_id TINYINT
last_update TIME...
Indexes
Data required to run the business
Inventory
film
film_id SMALLINT
title VARCHAR(255)
description TEXT
release_year YEAR
language_id TINYINT
original_language_id...
rental_duration TIN...
rental_rate DECIMA...
length SMALLINT
replacement_cost D...
rating ENUM(...)
special_features SET...
last_update TIMEST...
Indexes
Triggers
film_cate...
film_id SMALLINT
category_id TINYI...
last_update TIME...
Indexes
category
category_id TINYI...
name VARCHAR(25)
last_update TIME...
Indexes
language
language_id TINYI...
name CHAR(20)
last_update TIME...
Indexes
actor
actor_id SMALLINT
first_name VARCH...
last_name VARCH...
last_update TIME...
Indexes
film_actor
actor_id SMALLINT
film_id SMALLINT
last_update TIME...
Indexes
inventory
inventory_id MEDI...
film_id SMALLINT
store_id TINYINT
last_update TIME...
Indexes
film_text
film_id SMALLINT
title VARCHAR(255)
description TEXT
Indexes
Movie database

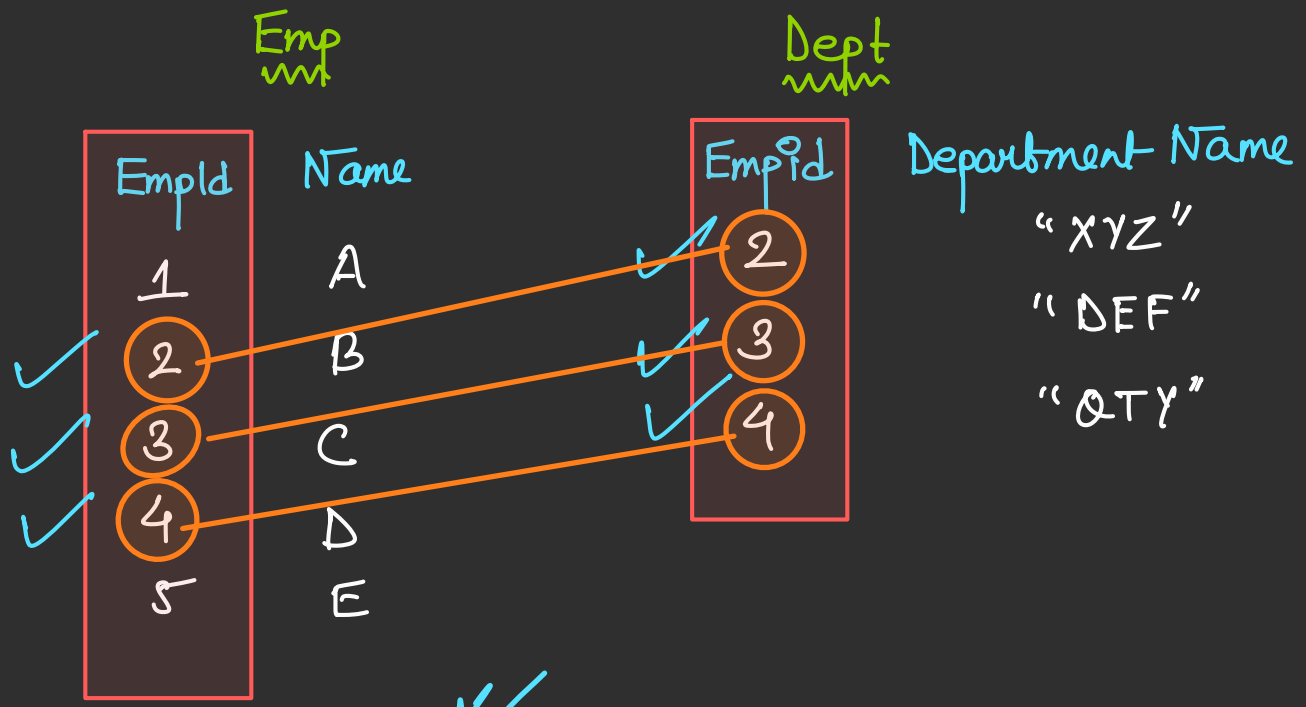
Table 1



Table 2



JOINS



"Common records b/w two tables"

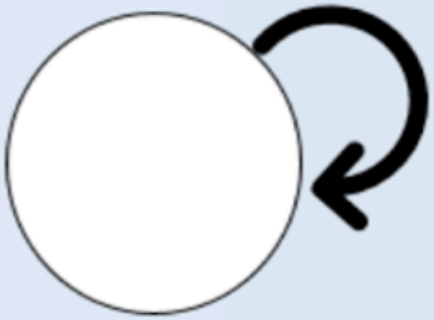
(INNER JOIN)

Query o/p

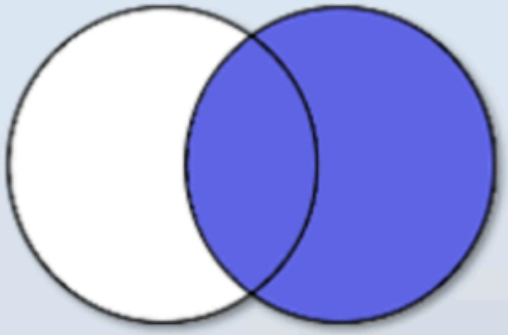
EmpId	Name	DeptNa
2	B	XYZ
3	C	DEF
4	D	QTY

Joins in MySQL

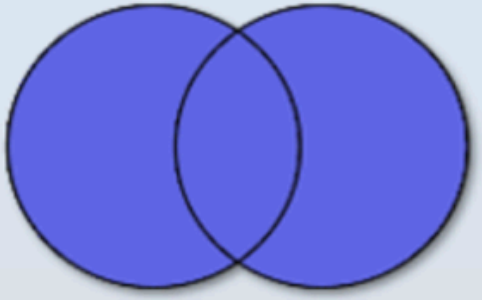
Self Join



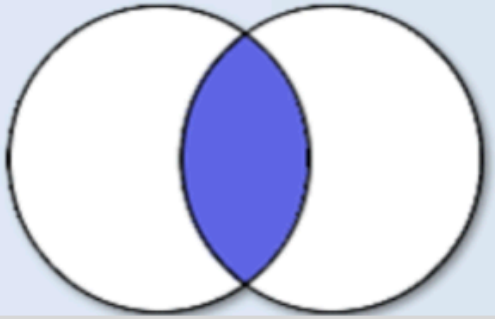
Right Join



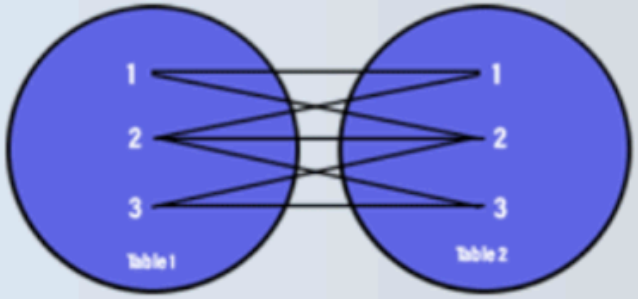
Full Join



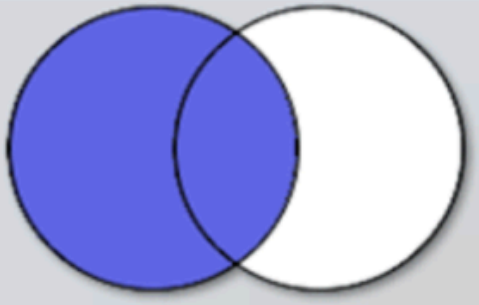
Inner Join

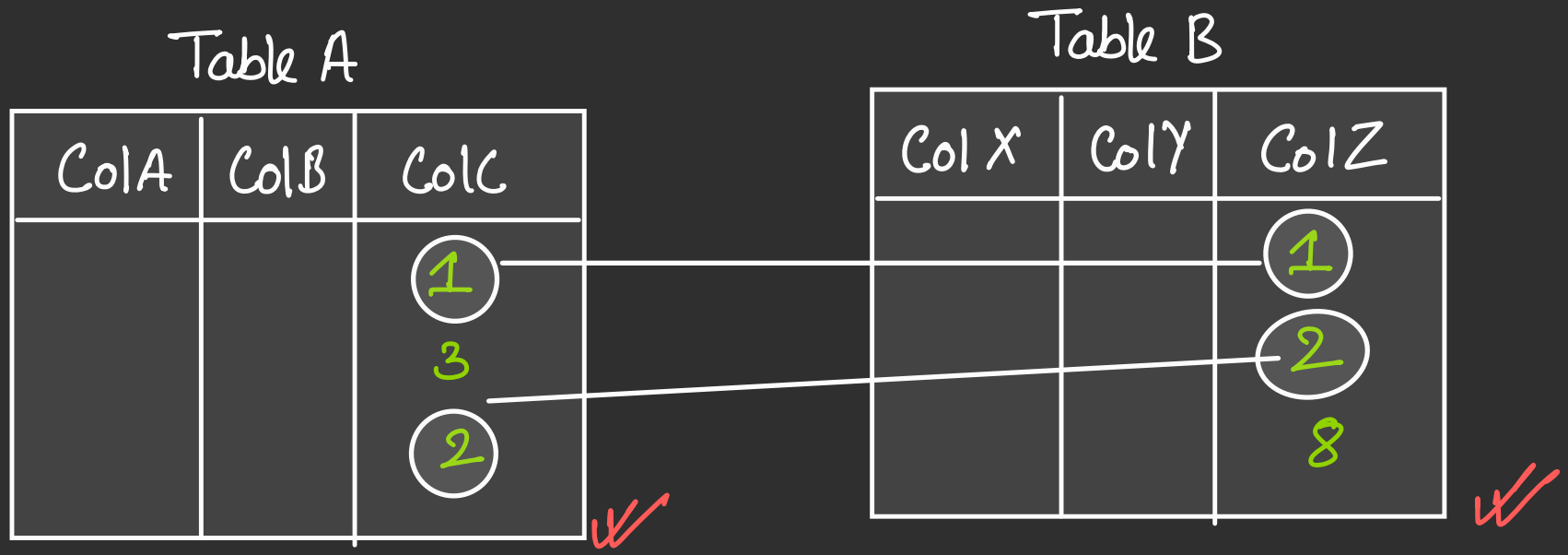


Cross Join



Left Join





perform Join → increasese/decreases no of rows

How many rows in o/p if Inner Join is done?

No of rows → 2

Q) How many records in final output?

Table A

ColA	ColB	ColC
Name	Add	1
		3
		2

Table B

ColX	ColY	ColZ
MNam	title	1
		2
		8

Inner Join

→ #2 (No of rows)

Table A

ColA	ColB	ColC
Name	Add	1
		3
		2

Table B

ColX	ColY	ColZ
MNbm	Hlk	1
		2
		2

Inner join → 3 records/rows

↳ All common records b/w two tables.

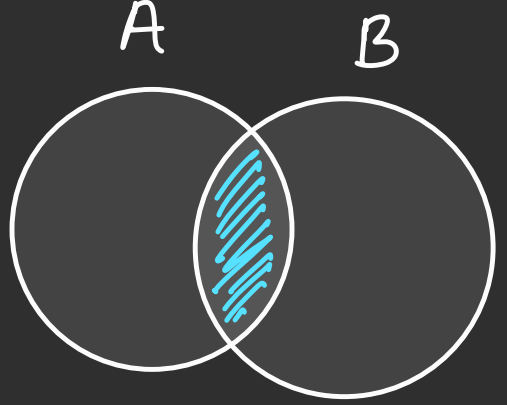


Table A

EmpID	Name
1	A
2	B

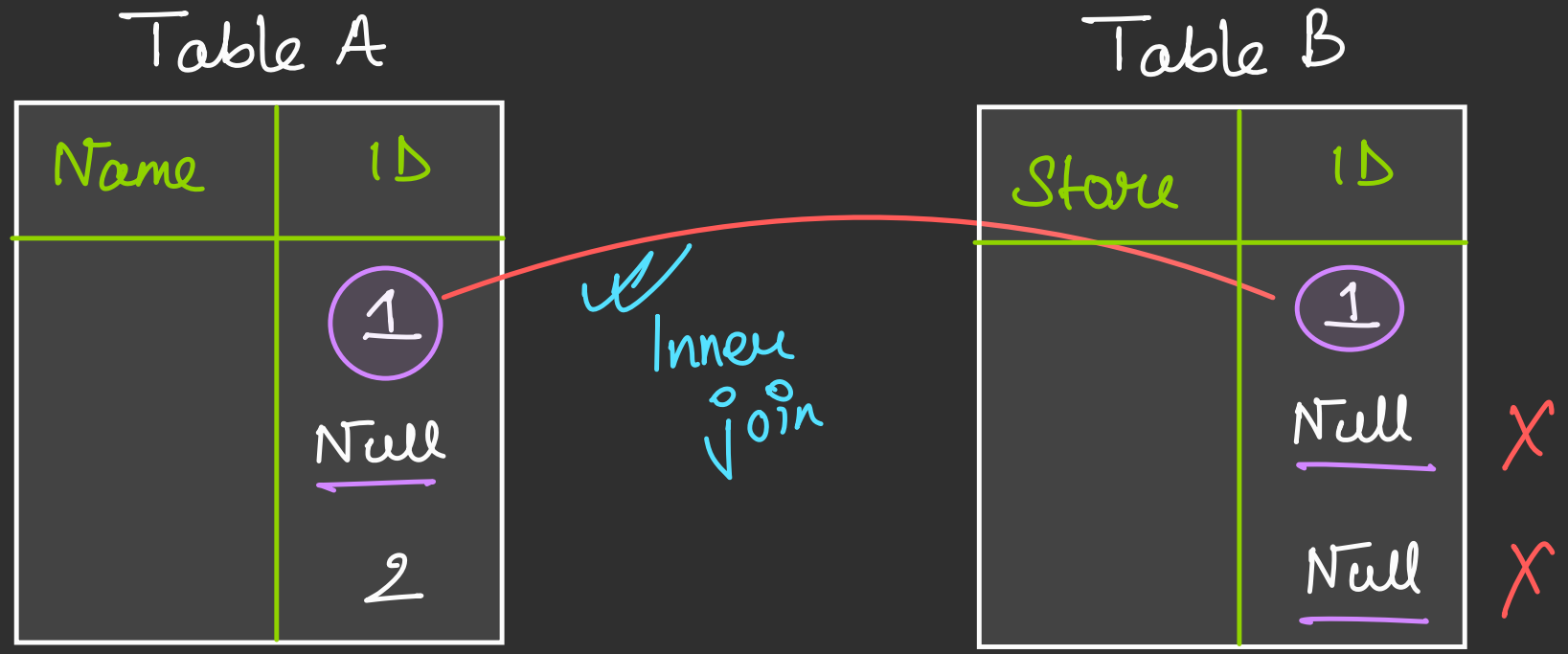
Table B

EmpID	DeptNam
1	X
1	Y

(Inner Join)

#2 records

EmpID	Name	EmpID	DeptNa
1	A	1	X
1	A	1	Y



No of rows → 1 row

Q) How many records in final output?

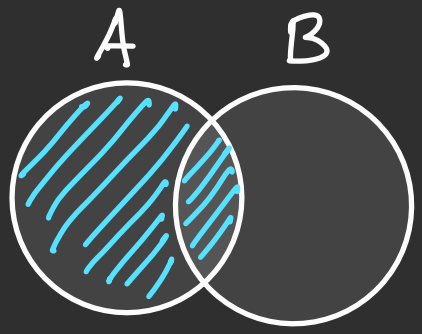
Table A

ColA	ColB	ColC
Name	Add	1
		3
		2

Table B

ColX	ColY	ColZ
MNam	title	1
		2
		8

Left Join →



ColC	ColZ
1	1
3	
2	2
—	8

3 records only

Q) How many records in final output?

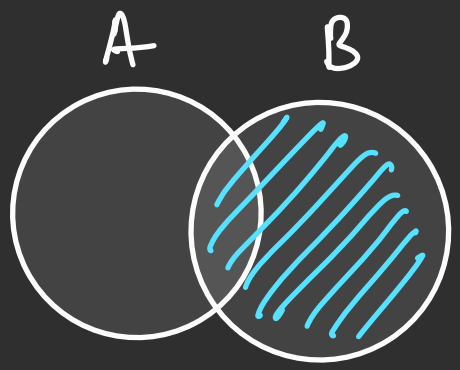
Table A

Col A	Col B	Col C
Name	Add	1
		3
		2

Table B

Col X	Col Y	Col Z
MNam	Hi Hi	1
		2
		8

Right Join →



3 records

Col A	Col B	Col C	Col X	Col Y	Col Z
—	—	1	—	—	1
—	—	2	—	—	2
Null	Null	Null	—	—	8

Q) How many records in final output?

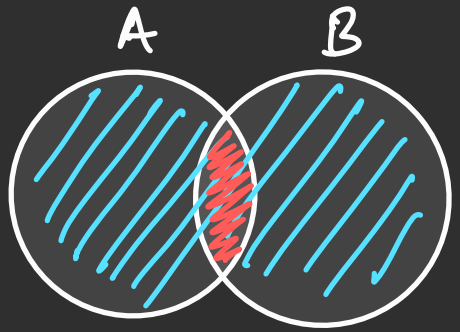
Table A

ColA	ColB	ColC
Name	Add	1
		3
		2

Table B

ColX	ColY	ColZ
MName	title	1
		2
		8

Full Outer Join:



1 — 1
3 —
2 — 2
— 8

#4 records



Q) What is the maximum no of records & minimum no of records possible in the following joins?

Homework

1) Inner Join → min, max

2) Left Join → min, max

3) Right Join → min, max

4) Full Outer Join → min, max

"general answer"

Left / Right

Full Outer

Inner

Self Join

emp Table (e1)

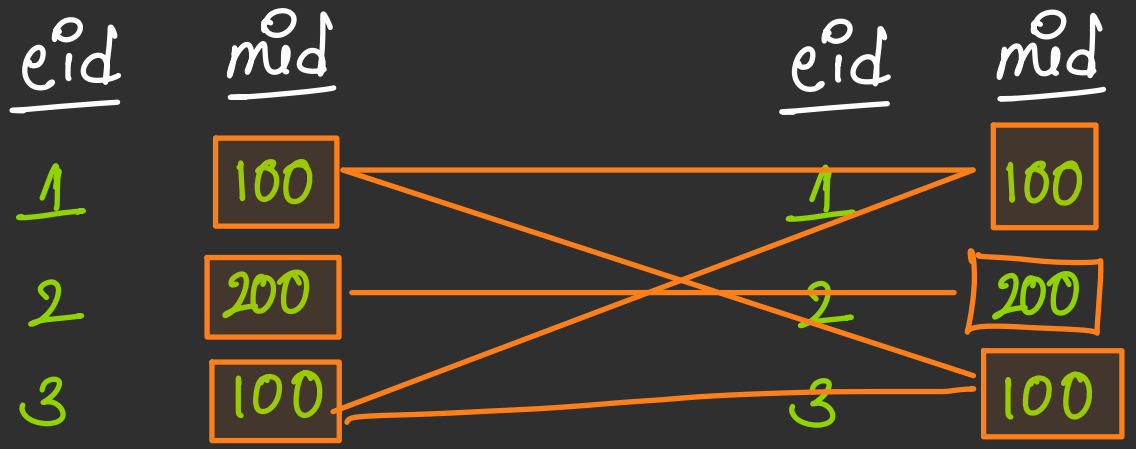
emp Table (e2)

Table 1

eid	mid
1	100
2	200
3	100

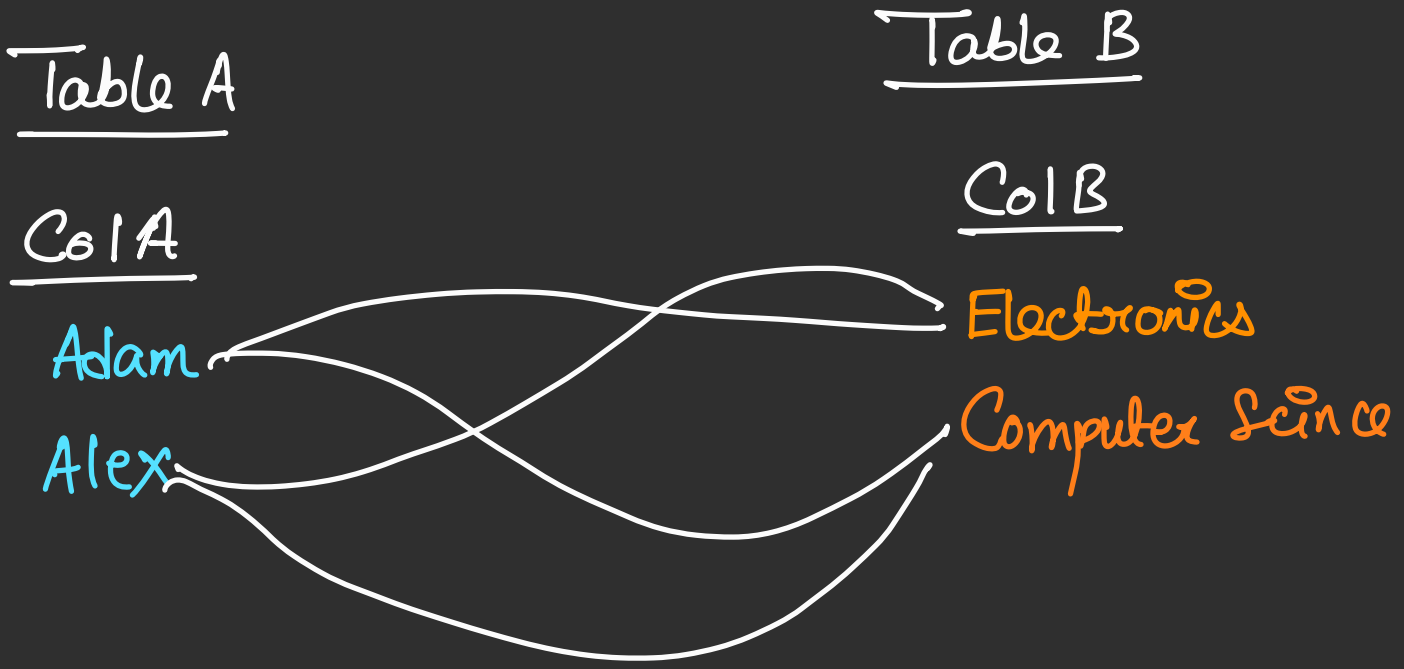
Table 2

eid	mid
1	100
2	200
3	100



eid	mid	eid	mid
1	100	1	100
1	100	3	100
2	200	2	200
3	100	1	100
3	100	3	100

Cross Join



ON Clause X

o/p → 2 x 2 = 4

Table ⇒ 30 X Table = 40 ⇒ 1200

Table ⇒ 1200 X Table = 1200 ⇒ 1440000

Table ⇒ 4 X Table = 3 ⇒ 4 x 3

12

Students

Student Informatio

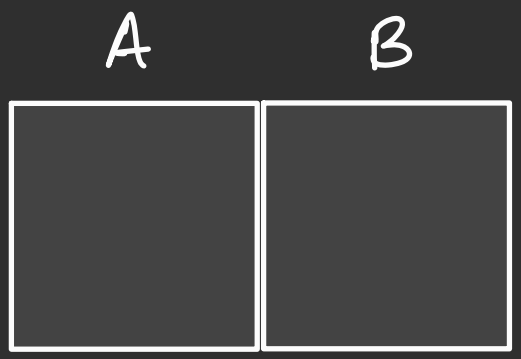
×

Subjects

Subject Informati.

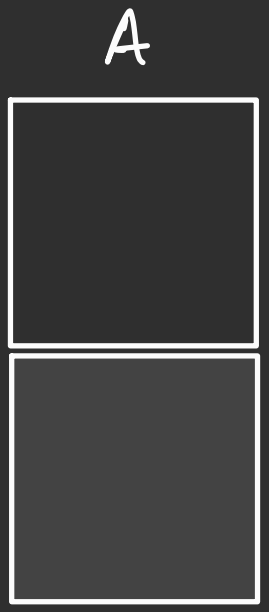
=>

All possible senario



Joins work

(Horizontal growth)



Union works

(Vertical growth)